HIMALAYAN EXPEDITIONS ANALYTICS

Power BI Project Report and Visual Insights

1. INTRODUCTION

The Himalayan Expeditions Analytics dashboard brings key expedition measures into one interactive report. Year, Season, and Gender slicers allow users to explore records across time, seasons, and member categories. The visuals examine expedition volume, participation patterns, recorded deaths, success rates, expedition duration, and oxygen-use behaviour. This consolidated view helps compare peaks and identify patterns in the dataset, provided that measures and aggregations are defined consistently.

2. DATA PREPARATION AND MODELLING

The expedition, member, and peak tables were imported into Power BI. Column names, data types, and table structures were reviewed before creating visuals. In Power Query, data quality was checked for missing values, errors, and inconsistent text. Category labels for nationality, gender, season, oxygen-use status, and expedition outcomes were standardised.
Duplicates were considered in relation to each table's level of detail. For example, repeated expedition IDs in member records can be valid because one expedition may include multiple climbers. Unknown values were treated separately from zero, and unusual values were retained unless there was a clear reason to classify them as errors.
The model was organised around Peaks, Expeditions, and Members, with relationships intended to support peak-level and member-level analysis. Measures were created for expedition totals, member totals, peak totals, success rate, recorded deaths, average expedition days, and oxygen-use rate. The report was explored using Year, Season, and Gender slicers. Member totals and death totals require careful interpretation: member records are not necessarily unique individuals, and counting nonblank death entries may not equal counting actual deaths.

3. OVERALL DASHBOARD AND KPI INSIGHTS

The dashboard combines headline indicators with visuals that describe expedition activity, participation, outcomes, and oxygen use. The main KPI values shown in the report are:

- Total expeditions: 11,421.
- Member records: 89,000.
- Recorded deaths: 1,158.
- Expedition success rate: 54.94%.
- Peaks represented: 480.

These figures summarise the scale and broad outcomes of the records included in the report. The success rate depends on the dashboard's definition of success and its denominator. The member figure should be treated as a record count unless a unique-person identifier has been validated. The death measure should also be checked to confirm that it counts actual deaths rather than simply nonblank values.



Figure 1. Dashboard and headline KPI overview from the original report.

4. DETAILED INSIGHTS FROM THE VISUALS

4.1 Expedition Trend by Year

- Recorded expedition activity is low in the early years and rises substantially after the mid-twentieth century, with a strong increase toward the late twentieth and early twenty-first centuries.
- Highest number of expeditions in 2014 and Lowest in 2020
- The sharp decline at the newest end of the trend may reflect incomplete recent-year records or limited reporting. Recent values should be interpreted cautiously until coverage is confirmed.

4.2 Climbers by Nationality and Gender

- Nepal is the largest nationality category by a clear margin in the displayed chart and USA represent more female climbers
- The UK, France, Spain, Germany, South Korea, Italy, and China appear as smaller groups in the displayed selection.
- The gender split provides a comparison of recorded participation across nationalities. These counts may represent member–expedition records rather than distinct individuals because one person may appear in multiple expeditions.

4.3 Recorded Deaths by Peak

- Everest has the highest displayed recorded death count, followed by Manaslu and Dhaulagiri I.
- Cho Oyu, Ama Dablam, and Lhotse have lower counts in the shown comparison.

4.4 Expeditions by Oxygen Use

- Approximately 8.05K records (70.45%) are shown as False and 3.38K records (29.55%) as True, out of 11,421 records.
- If the Boolean field indicates supplemental oxygen use, most displayed expedition records are categorised as not using oxygen.
- As the height increases the Oxygen rate increases , as there are more expeditions mid range height so Oxygen usage rate is low.

5. PEAK PERFORMANCE AND ELEVATION INSIGHTS

5.1 Peak-wise Expedition Performance

- Everest has 2,344 expeditions, a 63.10% success rate, 340 recorded deaths, and an average duration of 37.3 days in the displayed matrix.
- Ama Dablam has 1,576 expeditions and a 70.88% success rate.
- Annapurna I has 264 expeditions, a 36.36% success rate, and 73 recorded deaths, while Cho Oyu has 1,359 expeditions and a 61.81% success rate.
- The most frequently visited peak is not necessarily the peak with the highest success rate. Success rate, duration, and absolute death counts measure different aspects and should be considered together with sample size and exposure.

5.2 Expedition Distribution by Peak

- Everest has the largest displayed expedition volume at 2,344, followed by Ama Dablam at 1,576 and Cho Oyu at 1,359.
- Manaslu (775), Lhotse (512), Dhaulagiri I (424), Makalu (383), and Baruntse (335) appear as smaller groups in the treemap.
- Treemap rectangle size represents expedition volume only; it does not represent difficulty, popularity per year, success probability, or safety.

5.3 Oxygen Use Versus Expedition Height

- The scatter plot suggests that oxygen-use rates generally increase at higher elevations.
- Several of the tallest peaks have displayed rates above 50%, while many lower peaks are closer to the bottom of the chart.
- The pattern varies across peaks, and points at similar elevations can have different oxygen-use rates. Route, season, expedition strategy, and differences in the records may also affect oxygen use.
- Because the chart is observational, the relationship should be described as an association rather than proof that elevation alone causes oxygen use.

6. CONCLUSION

The Power BI dashboard brings Himalayan expedition activity, climber participation, peak-level outcomes, and oxygen-use patterns into one analytical report. The yearly trend indicates that recorded activity increased substantially across much of the historical period, especially in later decades, although the most recent decline may reflect incomplete records or limited reporting. Everest, Ama Dablam, and Cho Oyu account for comparatively large expedition volumes. Everest also has the highest displayed absolute death count, but this alone does not establish the greatest individual risk. Success rates vary across peaks, and expedition duration provides additional context when comparing outcomes. The oxygen-use visuals suggest a positive association between elevation and oxygen use, but the pattern varies and does not show that height is the only factor influencing oxygen decisions.

Before using the dashboard for stronger comparisons or decisions, validate the definitions and aggregation of member counts, deaths, success rates, and oxygen-use values, including the treatment of missing data. Comparing deaths with exposure measures such as expedition or climber totals would provide more meaningful context for safety analysis.